Marwadi University	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP (01CT0513)	**Aim:** Simulate smoothing and sharpening operations on images using frequency domain filters.	
Experiment-10	**Date:**	**Enrollment No:92400133128**

Aim: Simulate smoothing and sharpening operations on images using frequency domain filters.

Fourier Transform

The **Fourier Transform** converts an image from the spatial domain to the frequency domain. It represents the image using different frequency components.

- `np.fft.fft2(img)` – Converts the image into the frequency domain using the 2D Fourier Transform.
- `np.fft.fftshift()` – Moves the zero-frequency component to the center of the

3. THEORY

Frequency domain filtering is an image processing technique that uses the Fourier Transform to process images based on their frequency components.

Low-Pass Filter (LPF): It allows low-frequency components and removes high-frequency components. It is used to smooth images and reduce fine details and noise.

High-Pass Filter (HPF): It allows high-frequency components and removes low-frequency components. It is used to highlight edges and fine details for sharpening images.

Code:

```
import cv2

import numpy as np

import matplotlib.pyplot as plt


# 1. Read image using image path

img = cv2.imread(r'C:\Users\KOMARA MOHAN GOWDA\OneDrive\Desktop\My Studies\SEM-5\DSIP\Image Processing Assignemnt\Best_view.png', 0)


# 2. Fourier Transform

F = np.fft.fftshift(np.fft.fft2(img))

M, N = img.shape
```

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```
# 3. Create frequency distance matrix
u, v = np.meshgrid(np.arange(N) - N//2, np.arange(M) - M//2)
D = np.sqrt(u**2 + v**2)

# 4. Low-Pass and High-Pass Filters
LPF = D < 30
HPF = D >= 30

# 5. Apply filters
smooth = np.abs(np.fft.ifft2(np.fft.ifftshift(F * LPF)))
high_pass = np.abs(np.fft.ifft2(np.fft.ifftshift(F * HPF)))
sharp = np.clip(img.astype(float) + high_pass, 0, 255)

# 6. Display images
plt.figure(figsize=(12, 4))

plt.subplot(1, 3, 1)
plt.imshow(img, cmap='gray')
plt.title('Original Image')
plt.axis('off')
plt.subplot(1, 3, 2)
plt.imshow(smooth, cmap='gray')
plt.title('LPF - Smoothing')
plt.axis('off')
plt.subplot(1, 3, 3)
plt.imshow(sharp, cmap='gray')
plt.title('HPF - Sharpening')
plt.axis('off')
plt.show()
```

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View Go Run Terminal Help | Image Processing Assignemnt | Sign In

Welcome | Exp11.ipynb | Exp-10.ipynb | histogram.ipynb | images.jpeg | experiment5.ipynb | Best_view.png | Nice_view.png | histogram_ma

Exp-10.ipynb > import cv2

Generate | Code | Markdown | Run All | Restart | Clear All Outputs | Jupyter Variables | Outline

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```

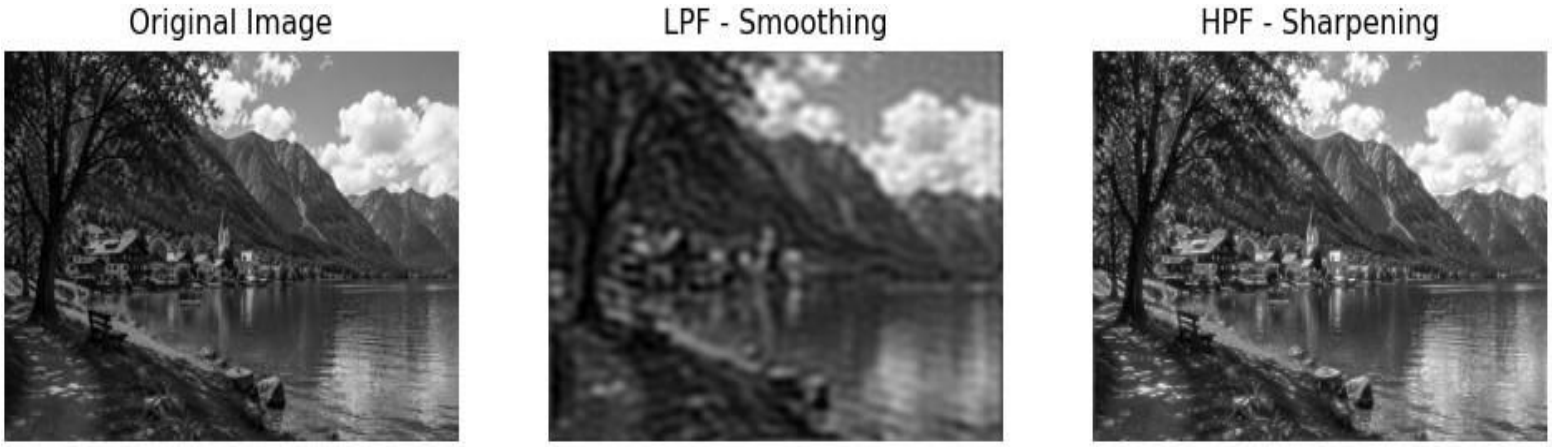
[10] ✓ 1.2s

Spaces: 4

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5. RESULT

The input image was processed using frequency domain filters. The Low-Pass Filter produced a smoothed image, while the High-Pass Filter highlighted fine details and edges to produce a sharpened image.

6. CONCLUSION

Thus, smoothing and sharpening operations were simulated successfully using Low-Pass and High-Pass Filters in the frequency domain.

Github: https://github.com/abhinayreddy-7/DSIP.git